Pitch Gym: AI-Powered Investor Simulation for Startup Pitch Training

School of Computer Science and Engineering – Winter Semester 2025 - 26

Introduction

Startup founders struggle to access structured, repeatable pitch training. Traditional methods like accelerator programs, human coaches, and peer feedback are infrequent, expensive, and subjective. Many new AI tools, such as ChatGPT, can engage startup entrepreneurs in simulated discussions with potential investors; however, these AI tools fall short in replicating the key element – voice. The delivery of pitches through voice, in terms of tone, confidence, pacing and spontaneity is a verbal communication skill that can only be evaluated through interaction via voice

Pitch Gym addresses this gap by combining real-time voice AI, structured scoring, and longitudinal progress tracking into a single web-based solution accessible to any founder with an internet connection.

Motivation

- Accelerator programs accept <3% of applicants
- Human pitch coaches cost $200–500/session
- Text-based AI tools cannot assess verbal delivery
- No existing platform offers voice-based pitch training with structured feedback

Scope of the Project

- 10 pitch scenarios (Elevator Pitch, VC Ready, Demo Day, Investor Skeptic Mode, etc.)
- Real-time voice conversations with AI investor personas
- Per-category scoring (1–5) with strengths, weaknesses, and suggestions
- Analytics dashboard with score trends and skill radar charts
- Context carry-forward between sessions for progressive coaching
- Credit-based monetization via Razorpay

Methodology

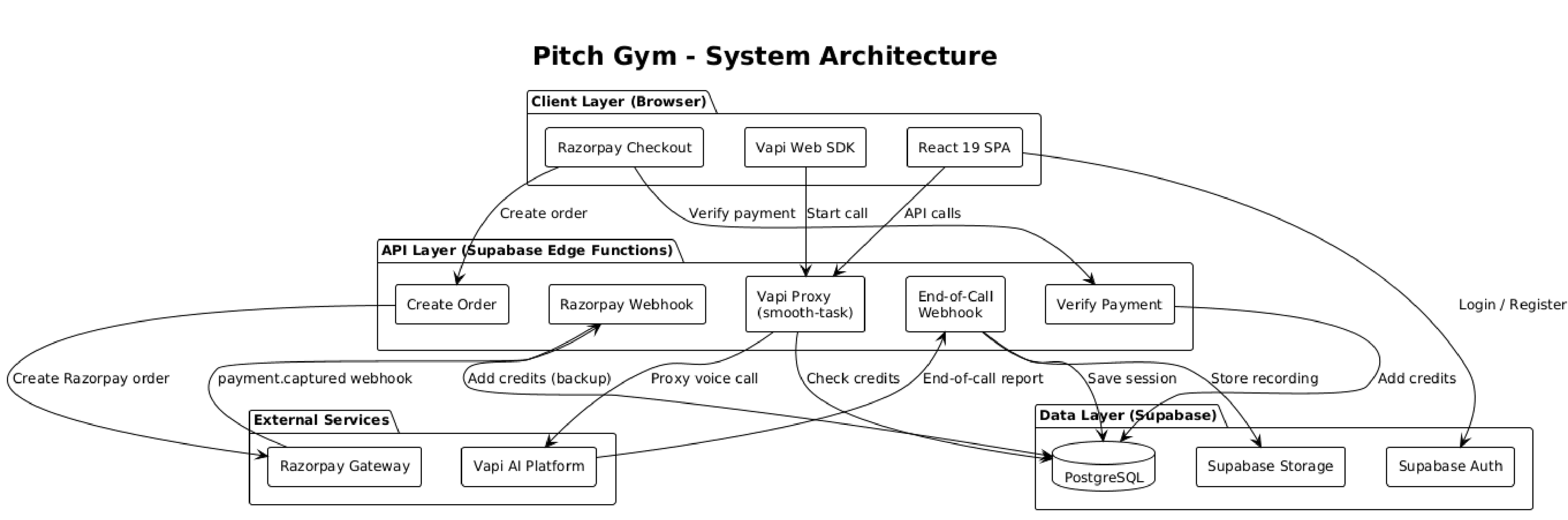
Stack: React 19 + Vite 7 | Supabase (PostgreSQL, Auth, Edge Functions) | Vapi Web SDK (WebRTC) | Razorpay | Vapi AI Agent - OpenAI GPT-4o (LLM) + Deepgram (STT) + ElevenLabs (TTS)

Key Design Decisions:

- 10 AI investor personas with specialized system prompts and structured output schemas
- Fractional credit system: users are charged per minute of actual call duration
- Dual-verification payment architecture (client-side + webhook) with idempotent credit addition
- Context carry-forward injects prior session feedback for progressive coaching

Flow:

- Founder selects pitch type, difficulty, and optional prior context
- Edge Function authenticates, checks credits, sets duration limits, proxies to Vapi API
- WebRTC audio session with live transcript and countdown timer
- Post-call webhook extracts structured assessment, stores recording, deducts credits
- Founder views scores, feedback, and tracks progress on analytics dashboard



Results

Scoring Behavior

The AI differentiated difficulty appropriately: easier types (Customer Discovery: 3.5/5, Co-Founder: 3.6/5) scored higher than harder ones (VC Ready: 2.8/5, Investor Skeptic: 2.5/5).

Context Carry-Forward

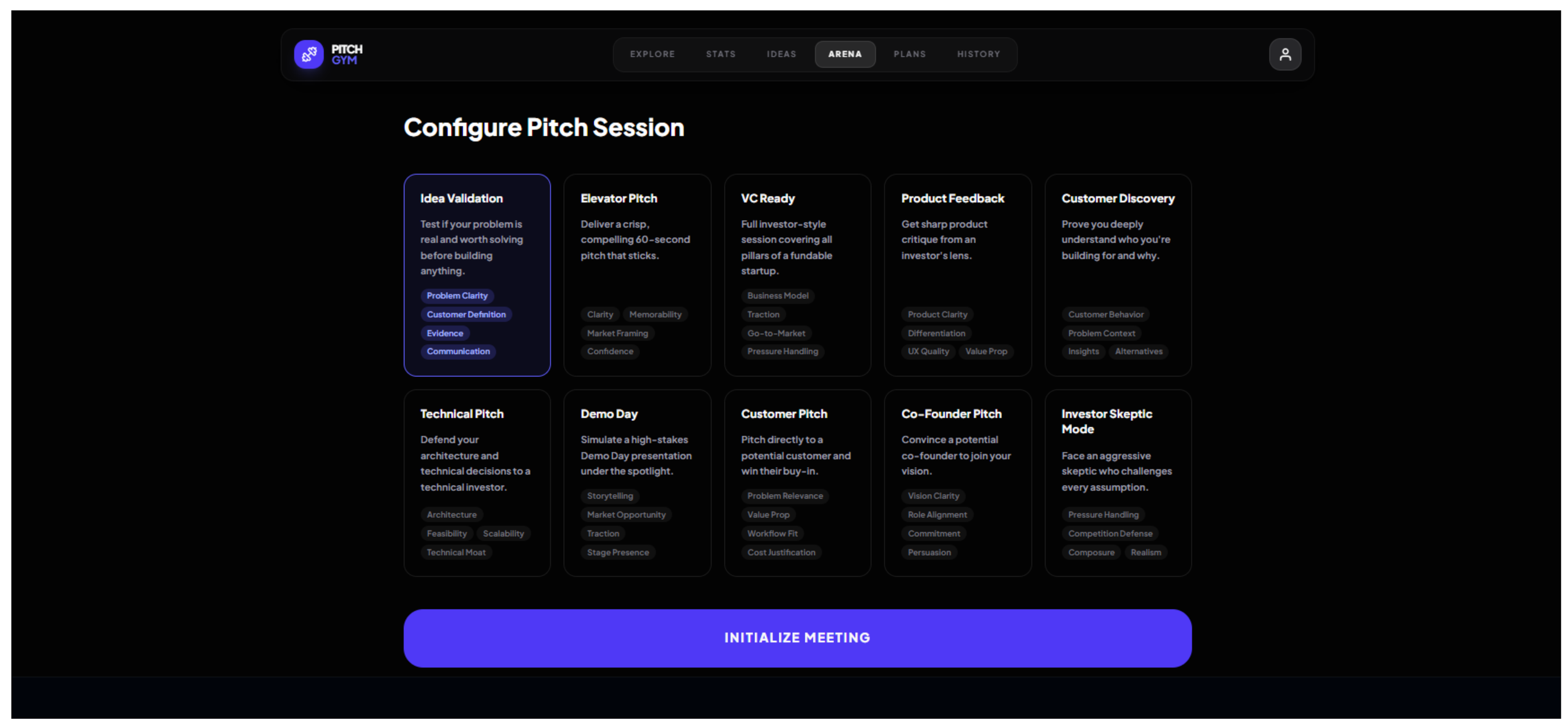
With prior context enabled, the AI acknowledged previous feedback and asked targeted follow-ups based on the founder's earlier performance.

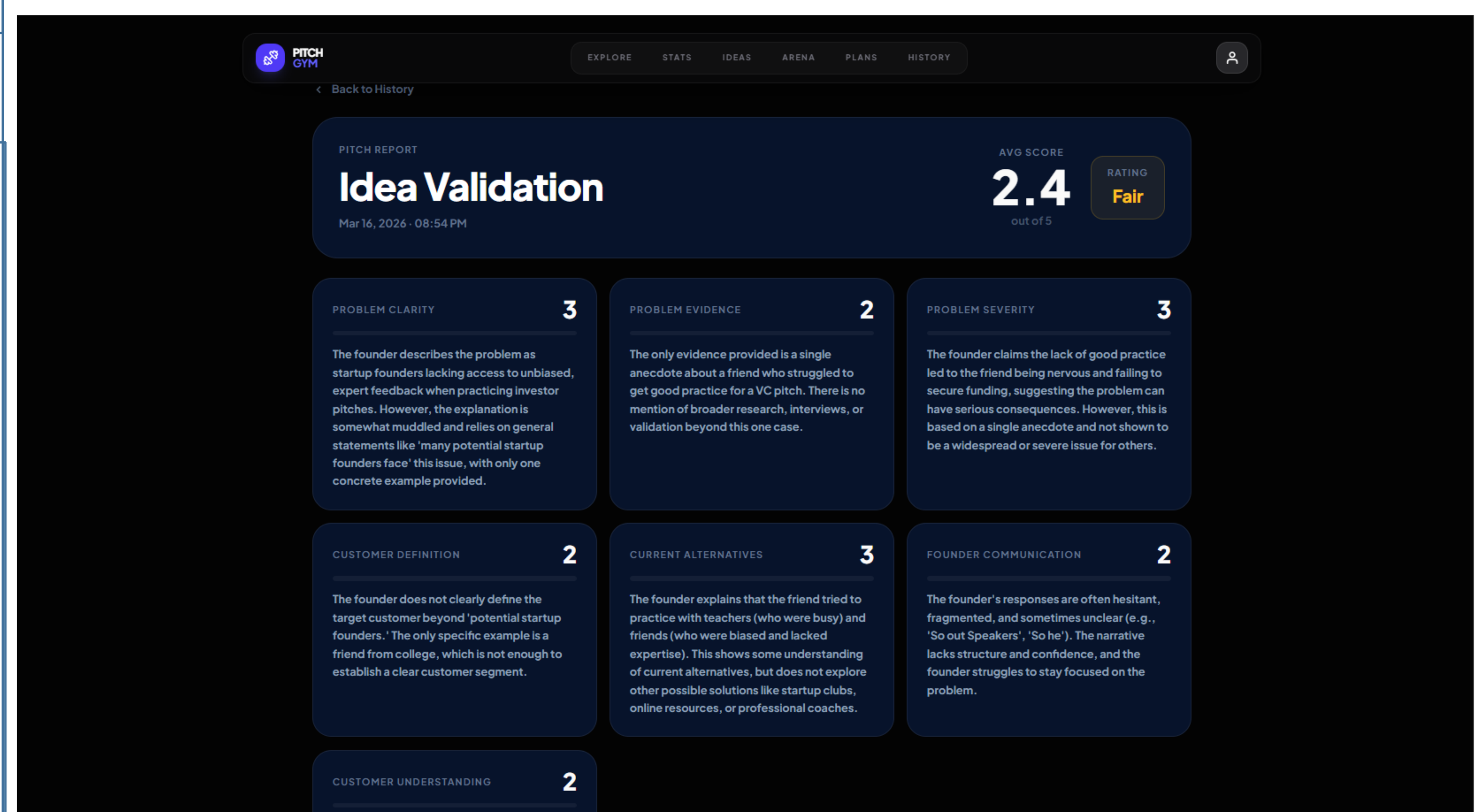
Testing

52 unit tests across pitchParser (32), pitchAnalytics (15), and pitchContext (5) using vitest: all passed in 1.30s. Integration and UAT validated OAuth, payments, webhooks, and end-to-end pitch flows across all 10 types.

Payment & Credit System

Dual-verification payment architecture processed all test transactions without double-crediting. Fractional credit deduction accurately tracked per-minute usage across sessions of varying lengths.





Conclusion

Pitch Gym demonstrates that voice AI can deliver structured pitch training previously only available through expensive human coaches. Assessments are consistent and the carry-forward mechanism enables progressive coaching.

Future work includes video-based delivery analysis and multi-language support.

References

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3. Radford et al. (2023). ICML.
4. Supabase Docs (2024). supabase.com/docs
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Demo Video Link:

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